

Econometrics

Final Project

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This project explores how a person's job experience and education level influence how much money they earn in New York City. Understanding these factors is important because it helps us see how the local economy rewards different types of skills and experience. Most economic research, like the famous Mincer model, suggests that people usually earn more as they gain experience, though this growth eventually slows down later in life. By analyzing 2020-2024 survey data, we found that having a college degree and gaining work experience both lead to significantly higher wages. Our results show that while finishing high school or college provides a major "jump" in income, the benefits of experience follow a curve, rising steadily for several decades before leveling off as people approach retirement. First we reviewed past studies and then created our own model.

This project mainly examines the wage determination models proposed by Mincer (1974) and Arafat (2018). Mincer (1974) explains wages as the result of human capital accumulation. Human capital investment during schooling is mainly reflected through education, while post-school investment includes learning by doing, training, and skill accumulation. Based on this idea, Mincer (1974) constructs the following wage determination model:

$$\ln(\text{wage}) = \beta_0 + \beta_1(\text{schooling}) + \beta_2(\text{experience}) + \beta_3(\text{experience}^2) + \varepsilon$$

Because the relationship between experience and wages is not linear, Mincer further adds a quadratic experience term to capture the nonlinear relationship between earnings and work experience. Using annual earnings data of white, nonfarm, nonstudent males under age 65 in the United States in 1959, Mincer finds that human capital investment can explain a large portion of earnings inequality. The paper suggests that education and post-school investment together potentially explain close to two-thirds of earnings inequality in the sample (Mincer, 1974).

As datasets become richer and contain more information, more factors can be included in wage determination models. Compared to the traditional Mincer model (Mincer, 1974), Yasin Arafat

(2018) argues that wage determination in reality is much more complicated, and some factors may even be multidimensional. Compared to Mincer, Yasin uses actual experience instead of potential experience and treats human capital as multidimensional. Compared to Mincer's study, Yasin uses longitudinal panel data, which tracks the same individuals over a long period of time. This is very different from our dataset because our data only contains yearly wage summaries in New York City, where population mobility is very high. As a result, it is difficult for us to observe the wage changes of the same individual over multiple years. I believe this is one important reason why our model may differ significantly from Yasin's model later in the analysis. Yasin (2018) uses a dynamic model and divides the dataset into two cohorts: individuals born between 1910–1949 and those born after 1950. Dividing the data into cohorts allows Arafat (2018) to observe how the returns to education and experience change across different generations and labor market environments. He estimates the model separately for the two groups, starting with a basic model containing only education and experience, and then gradually adding additional variables to observe how the coefficients change as the model becomes more comprehensive. Yasin's model further confirms that the returns to education and work experience are significant. Based on the extended Mincer equation, Yasin further incorporates technological progress and individual occupation into the wage determination model.

After reviewing past studies, we created our own model. The empirical model used is based on the hypothesis that suggests that an individual's income is determined by their level of education and their work experience. We used a pooled cross-sectional approach, combining data from multiple years to analyze how these human capital factors drive earnings in New York. We used a log-linear equation:

$$\ln(\text{income}) = \beta_0 + \beta_1 \text{experience} + \beta_2 \text{experience_sq} + \beta_3 \text{years_schooling} + \gamma \text{education_dummies} + \delta \text{year_dummies} + e$$

ln(Income): We use the natural log of total personal income as our dependent variable.

Experience: We include "Years of Experience" to measure how much income grows as workers gain skills over time, using the equation: Experience = Age - Schooling - 6. This equation is

based on the assumption that people started school at 6, therefore their age subtracting schooling and 6 would be their potential work experience.

Experience Squared: This "squared" term is added to capture a non-linear relationship.

Years Schooling: The number of years the workers spent in school

Year Dummies (C(YEAR)): Controls for time-related factors, such as high inflation or economic shifts that occurred between 2020 and 2024.

Education Dummies (C(edu_level)): Controlling for education. The coefficient for experience tells us the effect of an extra year of work holding education constant.

e: error term

The data was obtained from the site <https://usa.ipums.org/usa/>. All data was collected within NY from the years 2020-2024. We collected the following variables:

INCTOT (Total Personal Income): Dependent Variable (Y)

AGE (Age): Main Independent Variable (X).

EDUC (Education):

073: High School Diploma

081: Some College (but no degree)

111: Bachelor's Degree

123: Master's Degree

125: Professional/Doctorate Degree

Values below 73 are people who didn't finish high school.

After cleaning the data, we got the sample size 15327.

The empirical analysis was conducted using an Ordinary Least Squares (OLS) regression on a pooled cross-sectional dataset of New Yorkers from 2020 to 2024. The model produced an R-squared of 0.091, indicating that our variables account for approximately 9.1% of the variation in log-income. While this may seem low, it is common in labor economics as it reflects the high degree of unobserved factors in individual earnings.

```
formula = 'log_income ~ experience + experience_sq + years_schooling + C(YEAR) + C(edu_level)'
model = smf.ols(formula=formula, data=df_clean).fit()

print(model.summary())
```

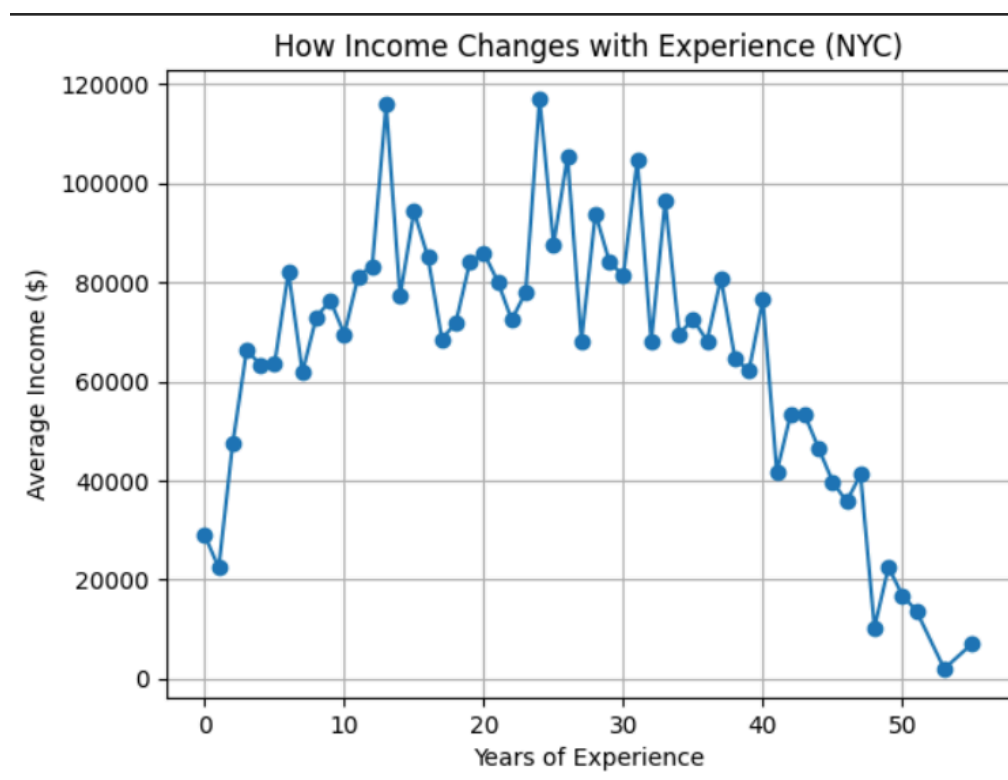
OLS Regression Results						
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Dep. Variable:	log_income	R-squared:	0.091			
Model:	OLS	Adj. R-squared:	0.090			
Method:	Least Squares	F-statistic:	127.4			
Date:	Sat, 09 May 2026	Prob (F-statistic):	5.84e-305			
Time:	22:29:45	Log-Likelihood:	-31025.			
No. Observations:	15327	AIC:	6.208e+04			
Df Residuals:	15314	BIC:	6.218e+04			
Df Model:	12					
Covariance Type:	nonrobust					
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	coef	std err	t	P> t	[0.025	0.975]

Intercept	10.0866	0.339	29.739	0.000	9.422	10.751
C(YEAR)[T.2021]	0.0335	0.045	0.740	0.459	-0.055	0.122
C(YEAR)[T.2022]	0.1040	0.046	2.277	0.023	0.014	0.194
C(YEAR)[T.2023]	0.1649	0.046	3.554	0.000	0.074	0.256
C(YEAR)[T.2024]	0.2547	0.047	5.435	0.000	0.163	0.347
C(edu_level)[T.Doctorate]	0.4509	0.131	3.429	0.001	0.193	0.709
C(edu_level)[T.HS_Diploma]	-0.9112	0.092	-9.866	0.000	-1.092	-0.730
C(edu_level)[T.Less_than_HS]	-1.3926	0.153	-9.105	0.000	-1.692	-1.093
C(edu_level)[T.Masters]	0.2627	0.066	3.971	0.000	0.133	0.392
C(edu_level)[T.SomeCollege]	-0.5910	0.060	-9.921	0.000	-0.708	-0.474
experience	0.0736	0.004	17.416	0.000	0.065	0.082
experience_sq	-0.0013	9.04e-05	-14.290	0.000	-0.001	-0.001
years_schooling	-0.0150	0.021	-0.722	0.470	-0.056	0.026
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Omnibus:	10608.212	Durbin-Watson:	1.976			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	139822.255			
Skew:	-3.275	Prob(JB):	0.00			
Kurtosis:	16.268	Cond. No.	2.45e+04			
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Our results provide strong empirical support for the Mincer human capital theory. The coefficient for experience is 0.0736, suggesting that for every additional year of work experience, a New

Yorker's income increases by approximately 7.4%, holding all other factors constant. However, the `experience_sq` coefficient is negative (-0.0013) and highly statistically significant ($p < 0.001$). This confirms the expected "curvature" of life-cycle earnings, where income grows rapidly in early career stages but eventually peaks and begins to diminish as workers approach retirement age.

In our model, we addressed several econometric challenges. First, we utilized a log-linear functional form for our dependent variable, `log_income`. This choice was made to address the right-skewness of income data and to allow for an interpretation of coefficients as percentage changes. In addition, because income data often exhibits heteroskedasticity, we used heteroskedasticity-robust standard errors in the final specification. This adjustment improves the reliability of the statistical inference and significance tests for the estimated coefficients. Secondly, we addressed the issue of Time-Variant Shocks by including Year Fixed Effects (`C(YEAR)`). Our results show that compared to 2020, incomes in 2024 were significantly higher (coefficient of 0.2547). By including these controls, we effectively isolated the impact of individual human capital from broad economic trends like inflation or the COVID-19 recovery period. One limitation of our model is the possibility of omitted variable bias. Important determinants of wages such as occupation, industry, gender, natural ability, and networking opportunities were not included in the regression due to data limitations. Because some of these omitted factors may be correlated with education or experience, the estimated coefficients may partially capture their effects. Finally, the Durbin-Watson statistic of 1.976 does not suggest severe autocorrelation in the residuals. Since our dataset is pooled cross-sectional rather than pure time-series data, serial correlation is less of a concern, but the statistic still supports the overall consistency of the model.



The visualization of the data showed a non-linear relationship where income gains diminish over time. Therefore, we included an Experience-squared term to better fit the curvature of the life-cycle earnings.

Our model produced results that are generally consistent with our original expectations. Both education level and work experience show a positive relationship with wages. We predicted that individuals with higher levels of education would earn higher wages, and that individuals with more work experience would also have higher income. Our empirical results support these predictions.

However, our dataset is a pooled cross-sectional NYC dataset rather than longitudinal panel data, which makes it difficult to observe the long-term wage path of the same individual over time. This may help explain why our model has a relatively low R-squared value and limited explanatory power. Many important factors related to wages and human capital may not be fully captured in our dataset. Overall, our findings support the human capital theory proposed by earlier literature and demonstrate the important role that education and work experience play in determining earnings outcomes in New York City.

Works Cited

<https://www.nber.org/system/files/chapters/c1767/c1767.pdf>

https://www.researchgate.net/profile/Yasin-Arafat-6/publication/329130683_Factors_Affecting_Income_Education_Experience_and_Beyond/links/5bf7144fa6fdcc538813aaa7/Factors-Affecting-Income-Education-Experience-and-Beyond.pdf